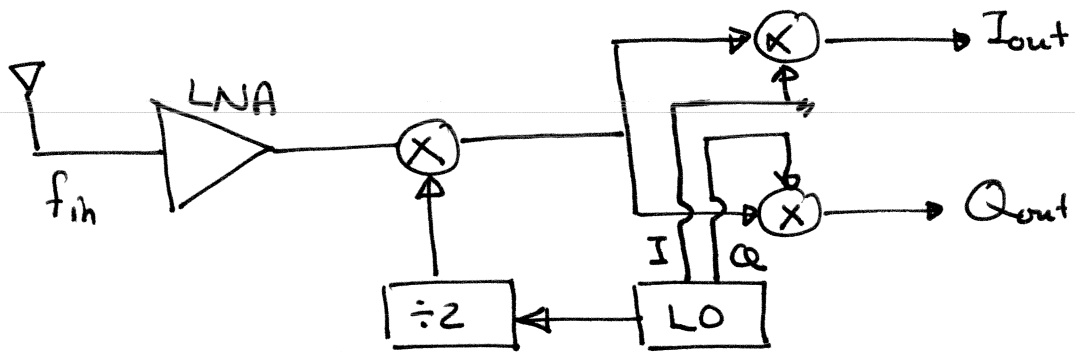


Problem 404:-



(a) Determine f_{ho} range as function of f_{in} .

$$f_{If_1} = f_{in} - f_{ho}/2.$$

$$f_{If_1} = f_{ho} \Rightarrow f_{in} = 1.5 f_{ho}.$$

for .11g $[f_{in} = 2.4 \text{ GHz} \rightarrow 2.485 \text{ GHz}]$.

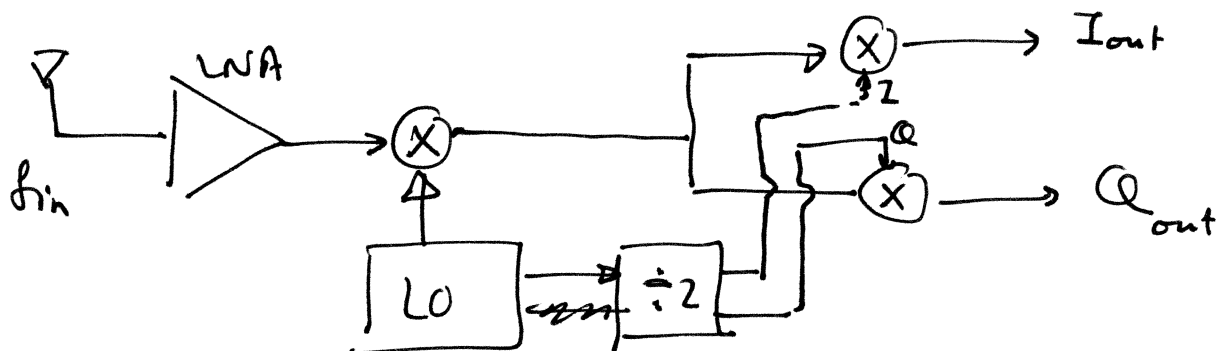
$$\Rightarrow f_{ho} = \frac{2}{3} f_{in} = 1.6 \text{ GHz} \rightarrow 1.657 \text{ GHz}.$$

(b) The image frequency range:

$$f_{im} = |f_{ho/2} - f_{If_1}| = |\frac{f_{ho}}{2} - f_{ho}| = \frac{f_{ho}}{2} \Rightarrow \text{Same as first mixer LO.}$$

$$= 0.8 \text{ GHz} \rightarrow 0.8285 \text{ GHz}.$$

(c)



Structure in fig. 4.26(a)

* Rx architecture of 4.26 (a) is better for many reasons -

1) Don't need to use quadrature Oscillator, as divide by 2 will give I & Q outputs directly.

2) Image frequency is @ $\frac{f_{LO}}{2}$. & first mixer LO is at f_{LO} so no results at dc.

while this one has output at $dc + 2(\frac{f_{LO}}{2}) = f_{LO}$.

→ Both of them has problem of Rf to If leakage at the image frequency.

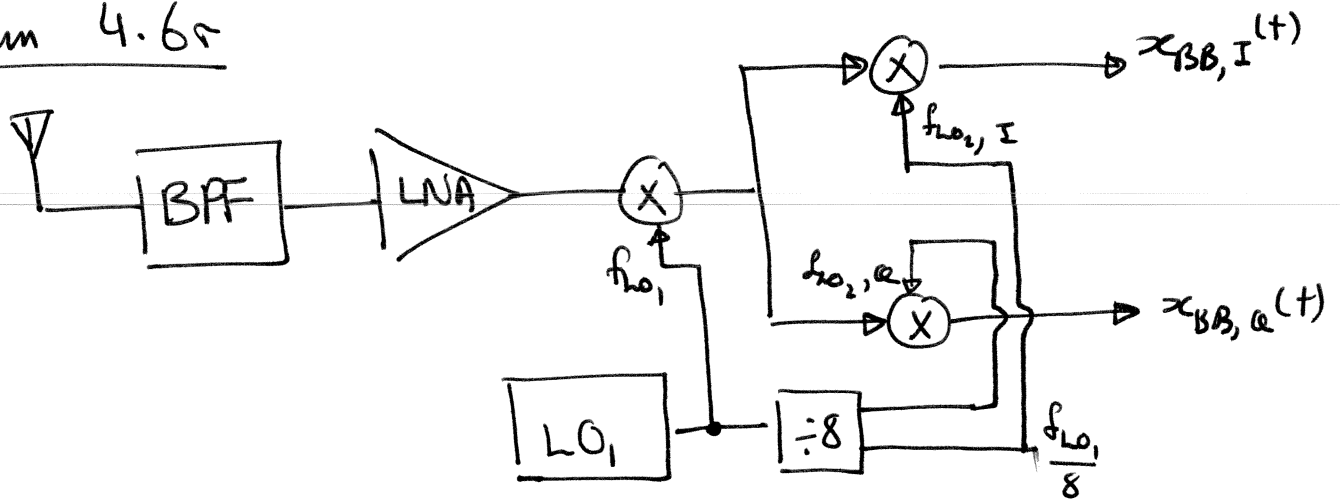
3) Matching of I & Q at a lower frequency is easier.

4) Lower Power Consumption.

5) flicker noise at input of first mixer will be upconverted to f_{If} . → not the case in 4.26 (a).

→ this will destroy the sensitivity of Rx.

Problem 4.6r



Sliding IF receiver for .11a. (802.11a) $\left\{ \begin{matrix} 5.25 - 5.725 \text{ GHz} \\ f_{in} \end{matrix} \right.$

(a) Image frequency Band:-

$$f_{IF_1} = f_{LO_1}/8 = f_{in} - f_{LO_1} \quad \Rightarrow \quad f_{in} = \frac{9}{8} f_{LO_1} \rightarrow f_{LO_1} = \frac{8}{9} f_{in}$$

$$f_{im} = f_{LO_1} - f_{IF_1} = f_{LO_1} - \frac{1}{8} f_{LO_1} = \frac{7}{8} f_{LO_1} = \frac{7}{9} f_{in}$$

$$f_{LO_1} = 4.66 \text{ GHz} \rightarrow 5.089 \text{ GHz}, \quad f_{im} = 4.083 \text{ GHz} \rightarrow 4.452 \text{ GHz}$$

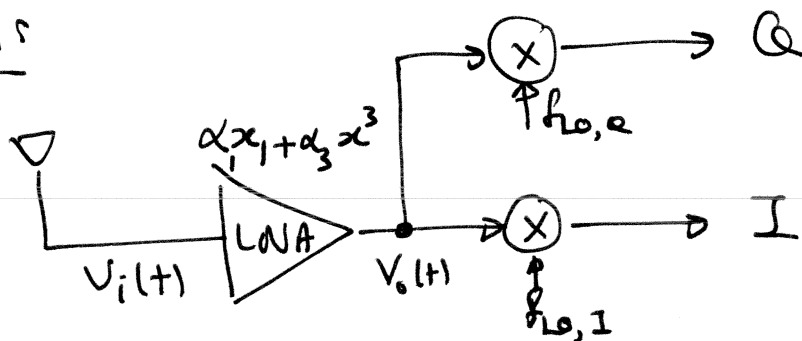
(b) because Mixing action is not ideal, there is problem of mixing with $3f_{LO}, 5f_{LO} \dots$

This can cause interferers to be converted to the desired frequency band.

$$\text{for first mixer: } f_{in,i} = 3f_{LO_1} \pm f_{IF_1} = \frac{23}{8} \text{ and } \frac{25}{8} f_{LO_1}$$

$$\text{for second mixer: } f_{in,i} = \pm 3f_{LO_2} \pm f_{LO_1} = \frac{5}{8} \text{ and } \frac{11}{8} f_{LO_1}$$

Problem 4.9



$$f_{\text{leakage}} = V_o \cos(\omega_{lo} t).$$

$$V_i(t) = V_o \cos(\omega_{lo} t) + V_{in}(t) \cos(\omega_{int} t).$$

$$\begin{aligned} (a) \quad V_o(t) &= \alpha_1 V_i + \alpha_3 V_i^3 \\ &= \alpha_1 V_o \cos(\omega_{lo} t) + \alpha_1 V_{in}(t) \cos(\omega_{int} t) \\ &\quad + \alpha_3 \left[V_o \cos(\omega_{lo} t) + V_{in}(t) \cos(\omega_{int} t) \right]^3 \\ &= \alpha_1 V_o \cos(\omega_{lo} t) + \alpha_1 V_{in}(t) \cos(\omega_{int} t) + \alpha_3 V_o^3 \cos^3(\omega_{lo} t) \\ &\quad + \alpha_3 3 V_o^2 \cos^2(\omega_{lo} t) V_{in}(t) \cos(\omega_{int} t) + 3 \alpha_3 V_o \cos(\omega_{lo} t) V_{in}^2(t) \cos^2(\omega_{int} t) \\ &\quad + \alpha_3 V_{in}^3(t) \cos^3(\omega_{int} t). \end{aligned}$$

near carrier

$$\begin{aligned} V_o(t) &= \alpha_1 V_o \cos(\omega_{lo} t) + \alpha_1 V_{in}(t) \cos(\omega_{int} t) + \alpha_3 \frac{V_o^3}{4} \times 3 \cos(\omega_{lo} t) \\ &\quad + \alpha_3 V_{in}^3(t) \times \frac{3}{4} \cos(\omega_{int} t) + \alpha_3 \frac{3V_o^2}{2} V_{in}(t) \cos(\omega_{int} t) \\ &\quad + \alpha_3 \frac{3V_o^2}{4} V_{in}(t) \cos[(2\omega_{lo} - \omega_{int})t] + \alpha_3 \frac{3V_o}{2} V_{in}^2(t) \cos(\omega_{lo} t) \\ &\quad + \alpha_3 \frac{3V_o}{4} V_{in}^2(t) \cos[(2\omega_{int} - \omega_{lo})t]. \end{aligned}$$

$$\begin{aligned}
 V_o(t) = & \cos[\omega_{lo}t] \left\{ \alpha_1 V_o + \alpha_3 \frac{3V_o^3}{4} + \alpha_3 \frac{3V_o}{2} U_{in}^2(t) \right\} \\
 & + \cos[\omega_{in}t] \left\{ \alpha_1 U_{in}(t) + \alpha_3 \frac{3}{4} U_{in}^3(t) + \alpha_3 \frac{3V_o^2}{2} U_{in}(t) \right\} \\
 & + \cos[(2\omega_{lo} - \omega_{in})t] \alpha_3 \frac{3V_o^2}{4} U_{in}(t) + \cos[(2\omega_{in} - \omega_{lo})t] \alpha_3 \frac{3V_o}{4} U_{in}^2(t)
 \end{aligned}$$

(b) baseband components

$$U_{o,BB}(t) = \left[\alpha_1 V_o + \alpha_3 \frac{3V_o^2}{4} + \alpha_3 \frac{3V_o}{2} U_{in}^2(t) \right] \rightarrow \text{dc}$$

$$+ \cos[(\omega_{in} - \omega_{lo})t] \left\{ \alpha_1 U_{in}(t) + \frac{3}{4} \alpha_3 U_{in}^3(t) + \frac{3}{2} V_o^2 U_{in}(t) \alpha_3 \right\}$$

$$+ \cos[(\omega_{lo} - \omega_{in})t] \alpha_3 \frac{3V_o^2}{4} U_{in}(t) + \cos[2(\omega_{in} - \omega_{lo})t] \alpha_3 \frac{3V_o}{4} U_{in}^2(t)$$

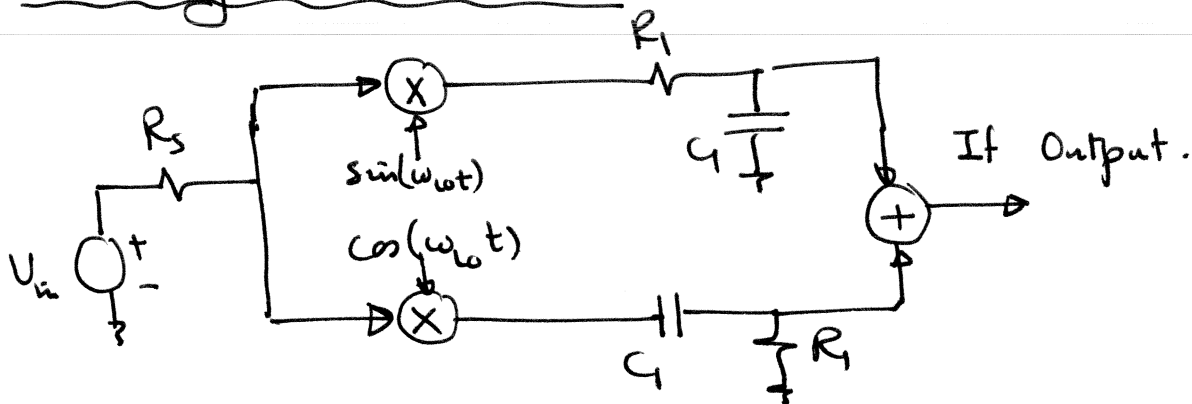
$$\text{signal} \sim \cos[(\omega_{in} - \omega_{lo})t] \left\{ \alpha_1 + \alpha_3 \frac{3}{2} V_o^2 + \alpha_3 \frac{3}{4} V_o^2 \right\} U_{in}(t)$$

$$\text{2nd order distortion} \quad \cos[2(\omega_{in} - \omega_{lo})t] \alpha_3 \frac{3V_o}{4} U_{in}^2(t)$$

$$\text{AM distortion} \quad \cos[(\omega_{in} - \omega_{lo})t] \alpha_3 \frac{3V_o}{4} U_{in}^3(t) + \alpha_3 \frac{3V_o}{2} U_{in}^2(t)$$

Problem 4.17r

Hartley Architectures



$\frac{\pi}{2}$ phase shift with equal amplitude at $\omega_{If} = \frac{1}{R_1 C_1}$.

as amplitude is function of frequency.

$$H(s) = \frac{-s C_1 R_1}{1 + s C_1 R_1} + \frac{j}{1 + s C_1 R_1} \rightarrow \pi/2 \text{ due to mixing.}$$

$$= \frac{j - s C_1 R_1}{1 + s C_1 R_1}$$

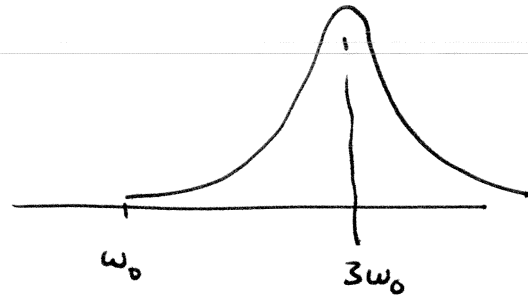
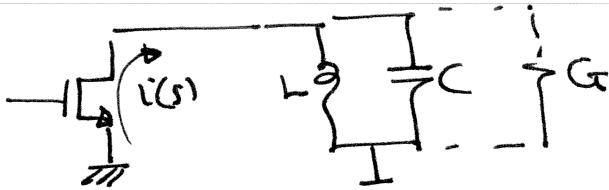
$$H(\omega_{If} + \Delta\omega) = \frac{j - j(\omega_{If} + \Delta\omega) R_1 C_1}{1 + j(\omega_{If} + \Delta\omega) R_1 C_1} = \frac{-j \Delta\omega R_1 C_1}{1 + j + j \Delta\omega R_1 C_1}$$

$$\approx \frac{-j \Delta\omega}{\omega_{If} (1 + j)}$$

$$IRR = \frac{|H_{signal}|^2}{|H(\omega_{If} + \Delta\omega)|^2} = \frac{2 \omega_{If}^2 |H_{signal}|^2}{\Delta\omega^2} = \frac{4 \omega_{If}^2}{(\Delta\omega)^2}$$

$$H_{signal}(s) = \frac{j + s C_1 R_1}{1 + s C_1 R_1} = \frac{2j}{1 + j} \Rightarrow |H_{signal}(\omega)| = \sqrt{2}$$

Problem 2.24



$$Q = \frac{\omega_{res} C}{G} = \frac{3\omega_0 C}{G} = \frac{\omega_{res} C}{\omega_{res}^2 L C G} = \frac{1}{\omega_{res} L G}$$

$$U_0(s) = i(s) \frac{1}{sC + G + \frac{1}{sL}}$$

$$= i(s) \frac{sL}{1 + sLG + s^2LC}$$

$$= i(s) \frac{s/\omega_{res} Q}{1 + s/\omega_{res} Q + s^2/\omega_{res}^2} = \frac{i(s)}{G} \frac{s/\omega_{res} Q}{1 + \frac{s}{\omega_{res} Q} + \frac{s^2}{\omega_{res}^2}}$$

$$|H(3\omega_0)|^2 = \left| \frac{i(3\omega_0)}{G} \right|^2 \left| \frac{j \frac{3\omega_0}{3\omega_0 Q}}{j \frac{3\omega_0}{3\omega_0 Q}} \right|^2 = \left| \frac{i(3\omega_0)}{G} \right|^2$$

$$|H(\omega_0)|^2 = \left| \frac{i(\omega_0)}{G} \right|^2 \left| \frac{j \omega_0 / 3\omega_0 Q}{1 - \frac{1}{9} + j \frac{\omega_0}{3\omega_0 Q}} \right|^2$$

$$= \left| \frac{i(\omega_0)}{G} \right|^2 \frac{(1/3Q)^2}{(8/9)^2 + (1/3Q)^2} \approx \left| \frac{i(\omega_0)}{G} \right|^2 \frac{(1/Q)^2}{(8/3)^2}$$

$$\frac{V_0(\omega)}{U_0(3\omega_0)} = \sqrt{\frac{|H(\omega_0)|^2}{|H(3\omega_0)|^2}} \approx \frac{3}{8Q} \rightarrow \text{Attenuation factor} \approx \frac{8Q}{3}$$